Drugs

Mark scheme and worked solutions for 20 questions: 10 multiple choice, 10 structured.

1 · Part 1

Core Drugs [1]

Answer: C

Explanation

Option C is the syllabus description of a drug: any substance taken into the body that modifies or affects chemical reactions in the body. Option A describes only some drugs, such as antibiotics, not all of them. Option B describes substances made by the body, such as enzymes or hormones, which are not taken in. Option D describes a nutrient in food.

2 · Part 1

Core Drugs [1]

Answer: A

Explanation

Antibiotics are drugs because they are taken into the body and affect chemical reactions there (statement 2), and the description of a drug says it is taken into the body (statement 3). Statement 1 is wrong because substances such as caffeine are drugs but are not prescribed medicines. Statement 4 is wrong because most drugs do not kill body cells; antibiotics kill bacteria, not human cells. So options B, C and D each include a wrong statement.

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3 · Part 1

Core Drugs [1]

Answer: D

Explanation

Antibiotics kill bacteria but do not affect viruses. Option A wrongly says viruses are killed. Option B reverses the effects. Option C ignores the fact that antibiotics are used to treat bacterial infections because they kill bacteria.

4 · Part 1 Core Drugs [1]

Answer: B

Explanation

Ravi's cold is caused by a virus, and antibiotics do not affect viruses, so the antibiotic cannot cure the cold. Option A is wrong because resistance to antibiotics is a problem in bacteria, and the antibiotic would have no effect on the virus anyway. Option C is wrong because antibiotics do not kill white blood cells. Option D is wrong because colds are transmissible: the virus passes from person to person.

5 · Part 1 Core Drugs [1]

Answer: A

Explanation

Penicillin, like other antibiotics, kills bacteria, so it can only cure an infection caused by bacteria that are not resistant to it. Patient P fits this. Patient Q has bacteria that are resistant, so penicillin will be much less effective. Patients R and S have viral infections, and antibiotics do not affect viruses.

6 · Part 1 Core Drugs [1]

Answer: C

Explanation

Resistant bacteria are not killed by the antibiotic, so the antibiotic becomes less effective at treating infections caused by them. Option A is wrong because antibiotics never affect viruses. Option B is the opposite of resistance. Option D is wrong because resistant bacteria are still pathogens and can still cause disease, which is why resistance is a problem.

7 · Part 1 Core Drugs [1]

Answer: B

Explanation

The clear area is where the antibiotic has killed the bacteria (or stopped them growing), so the widest clear area shows the most effective antibiotic. X has the largest diameter. W has no clear area, so these bacteria are resistant to W. Y and Z killed fewer bacteria, giving smaller clear areas.

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8 · Part 1 Extended Drugs [1]

Answer: D

Explanation

Each time antibiotics are used, non-resistant bacteria are killed and any resistant bacteria survive and reproduce, so resistant bacteria are selected. Using antibiotics only when essential means fewer occasions when this happens. Option A is wrong because mutations happen randomly; the antibiotic does not cause them. Option B is wrong because it is the bacteria, not the human body, that become resistant. Option C is wrong because resistant bacteria are not killed when antibiotics are absent; they simply have no advantage.

9 · Part 1 Extended Drugs [1]

Answer: C

Explanation

First a random mutation produces a resistant bacterium (2). The antibiotic then kills the non-resistant bacteria (3). The resistant bacteria survive and reproduce (1), passing on the gene until most of the population is resistant (4). Options A and D put the antibiotic before the mutation, suggesting the antibiotic causes resistance. Option B has resistant bacteria surviving and reproducing before the antibiotic is used, which removes the reason for their survival advantage.

10 · Part 1 Core Drugs [1]

Answer: A

Explanation

The percentage of resistant samples rose from 4% to 41%, and resistance reduces the effectiveness of an antibiotic, so antibiotic T was likely to be less effective in 2020. Option B is wrong because only 41% of samples were resistant in 2020. Option C is wrong because the increases are not doublings (for example 28% to 41% between 2015 and 2020). Option D is wrong because the table shows the percentage of resistant samples, not the number of patients infected.

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11 · Part 1 Core Drugs [4]

Mark scheme

(a) **[2]**

- any substance taken into the body (1)
- that modifies or affects chemical reactions in the body (1)

ALLOW changes / alters for modifies

REJECT substance made by the body, for the first mark

(b) **[1]**

- bacteria (1)

ALLOW bacterium / bacterial infections

REJECT viruses / any answer that includes viruses

(c) **[1]**

- antibiotics do not affect viruses (1)

ALLOW antibiotics only kill bacteria / antibiotics do not kill viruses

Worked solution

(a) The syllabus describes a drug as any substance taken into the body that modifies or affects chemical reactions in the body. You need both ideas: it is taken in from outside, and it changes chemical reactions. (b) Antibiotics are used to treat infections caused by bacteria. (c) Influenza is caused by a virus. Antibiotics kill bacteria but do not affect viruses, so an antibiotic cannot cure influenza.

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12 · Part 1 Core Drugs [5]

Mark scheme

(a) **[2]**

- the antibiotic killed the bacteria / the number of live bacteria in flask 1 decreased (1)
- data quote, e.g. from 2000 to 100 in 1 cm^3 over 8 hours, or comparison with flask 2 where the number rose to 32 000 (1)

ALLOW the number halved every 2 hours between 2 hours and 6 hours

IGNORE the antibiotic stopped bacteria growing, if the decrease is also described

(b) **[2]**

- (2000 − 100) ÷ 2000 × 100 or 1900 ÷ 2000 × 100 (1)
- 95 (%) (1)

ALLOW 95% with no working for 2 marks

(c) **[1]**

- as a control, to show that the decrease in flask 1 was caused by the antibiotic (1)

ALLOW to compare with / to show what happens to the bacteria without antibiotic

IGNORE to make it a fair test, on its own

Worked solution

(a) In flask 1, with the antibiotic, the number of live bacteria fell from 2000 to 100 in 1 cm^3 over 8 hours, so the antibiotic was killing the bacteria. In flask 2, with no antibiotic, the number doubled every 2 hours to 32 000. (b) The decrease is 2000 − 100 = 1900. Percentage decrease = 1900 ÷ 2000 × 100 = 95%. Check: 100 is 5% of 2000, and 100% − 5% = 95%. (c) Flask 2 is a control. It shows that without the antibiotic the bacteria increase, so the fall in flask 1 must be due to the antibiotic.

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13 · Part 1 Core Drugs [4]

Mark scheme

(a) **[1]**

- patient 1 (1)

(b) **[2]**

- influenza is caused by a virus (1)
- antibiotics do not affect viruses / antibiotics only kill bacteria (1)

IGNORE references to resistance

(c) **[1]**

- the new infection is caused by bacteria, which antibiotics can kill (1)

ALLOW antibiotics are used to treat bacterial infections

ALLOW HIV has weakened the immune system so the body cannot fight the bacterial infection without an antibiotic

REJECT the antibiotic will treat HIV

Worked solution

(a) Only patient 1 has an infection caused by bacteria. Antibiotics are used to treat bacterial infections. (b) Patient 2 has influenza, which is caused by a virus. Antibiotics kill bacteria but do not affect viruses, so it would not help. (c) HIV is a virus, so an antibiotic will not affect it. But the new lung infection is caused by bacteria, and antibiotics can kill bacteria, so an antibiotic is now useful to treat that bacterial infection.

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14 · Part 1 Core Drugs [5]

Mark scheme

(a) **[2]**

- 126 ÷ 200 × 100 (1)
- 63 (%) (1)

ALLOW 63% with no working for 2 marks

(b) **[2]**

- F (1)
- the most / highest number or percentage of samples were resistant to F, e.g. 126 out of 200 (1)

REJECT second mark if antibiotic E or G is named

(c) **[1]**

- G (1)

Worked solution

(a) Percentage resistant = 126 ÷ 200 × 100 = 63%. Check: 63% of 200 is 0.63 × 200 = 126. (b) Antibiotic F would be least effective because the bacteria in most of the samples (126 of 200) were resistant to it, and resistance reduces the effectiveness of an antibiotic. (c) Only 2 of the 200 samples were resistant to antibiotic G, so G is the most likely to kill the bacteria in a new patient.

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15 · Part 1 Core Drugs [6]

Mark scheme

(a) **[2]**

- 72 ÷ 80 × 100 (1)
- 90 (%) (1)

ALLOW 90% with no working for 2 marks

(b) **[2]**

- bacterial sore throats: the antibiotic increased the number or percentage that recovered, e.g. 72 compared with 36 / 90% compared with 45% / doubled (1)
- viral sore throats: the antibiotic had no effect, the same number (33) recovered with and without it (1)

(c) **[2]**

- antibiotics do not affect viruses (1)
- so the antibiotic gave no benefit / the same number recovered with or without it (1)

ALLOW patients who recovered did so without help from the antibiotic / because of their own immune system

IGNORE references to resistance

Worked solution

(a) Percentage recovered = 72 ÷ 80 × 100 = 90%. Check: 90% of 80 = 72. (b) For bacterial sore throats, 72 of 80 recovered with the antibiotic but only 36 of 80 without it (90% compared with 45%), so the antibiotic doubled the number who recovered. For viral sore throats, 33 of 60 recovered in both groups, so the antibiotic made no difference. (c) Antibiotics kill bacteria but do not affect viruses. The antibiotic could not act on the virus, so the same number of patients recovered whether or not they took it; those who got better did so without the antibiotic's help.

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16 · Part 1 Core Drugs [5]

Mark scheme

(a) **[2]**

- decreased from day 0 to day 4 / from 5000 to 400 (1)
- then increased from day 4 to day 8 / from 400 to 2400 (1)

ALLOW correct data quotes for either mark

(b) **[2]**

- some of the bacteria were resistant to the antibiotic / the antibiotic did not kill all of the bacteria because some were resistant (1)
- the resistant bacteria survived and reproduced / multiplied (1)

REJECT Maria became resistant to the antibiotic

(c) **[1]**

- the bacteria were resistant to the first antibiotic, so a different antibiotic is needed that can kill them (1)

ALLOW the first antibiotic was no longer effective

Worked solution

(a) The number of bacteria fell from 5000 on day 0 to 400 on day 4, then rose again to 2400 on day 8. (b) The early fall shows the antibiotic killed most bacteria. Since Maria kept taking it, the later rise suggests some bacteria were resistant. These were not killed, so they survived and reproduced, and the number of bacteria went up again. (c) Resistance reduces the effectiveness of the antibiotic, so the doctor chose a different antibiotic that these bacteria are not resistant to.

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17 · Part 1 Core Drugs [4]

Mark scheme

(a) **[1]**

- true (1)

(b) **[1]**

- false (1)

(c) **[1]**

- true (1)

(d) **[1]**

- false (1)

Worked solution

(a) True: antibiotics are used to treat bacterial infections. (b) False: antibiotics kill bacteria but do not affect viruses, so they cannot cure a cold. (c) True: some bacteria are resistant to antibiotics, which makes the antibiotics less effective. (d) False: a drug is a substance taken into the body that modifies or affects chemical reactions in the body; it is not made by the body.

18 · Part 1 Core Drugs [4]

Mark scheme

(a) **[1]**

- 16–30 (years) (1)

(b) **[1]**

- 152 (1)

(c) **[2]**

- antibiotics only kill bacteria / antibiotics kill bacteria (1)
- antibiotics do not affect viruses (1)

IGNORE references to resistance

Worked solution

(a) The highest value in the table is 38%, for the 16–30 group. (b) 38% of 400 = 0.38 × 400 = 152 people. Check: 152 ÷ 400 = 0.38. (c) Antibiotics kill bacteria, but they do not affect viruses. Colds are caused by viruses, so antibiotics cannot cure a cold.

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19 · Part 1 Extended Drugs [6]

Mark scheme

(a) **[3]**

- the antibiotic kills the non-resistant bacteria (1)
- the resistant bacteria survive and reproduce / multiply, passing on the gene for resistance to their offspring (1)
- the population becomes mostly resistant / the proportion of resistant bacteria increases (each time the antibiotic is used) (1)

REJECT the antibiotic causes the bacteria to become resistant

(b) **[2]**

- bacteria are exposed to antibiotics less often / fewer times (1)
- so there are fewer chances for resistant bacteria to survive while non-resistant bacteria are killed / less selection of resistant bacteria (1)

ALLOW resistant bacteria have less of an advantage, so they do not increase in number as quickly

REJECT patients become resistant to the antibiotics

(c) **[1]**

- MRSA bacteria are resistant to (several) antibiotics, so the antibiotics are less effective / do not kill them (1)

Worked solution

(a) When the antibiotic is used, it kills the bacteria that are not resistant. The few resistant bacteria survive and reproduce, passing on the gene for resistance to their offspring. Each time the antibiotic is used, this happens again, so the resistant bacteria make up more and more of the population. (b) If antibiotics are used only when they are essential, bacteria are exposed to them less often. There are fewer occasions when non-resistant bacteria are killed while resistant ones survive, so resistant strains such as MRSA are selected less often and develop more slowly. (c) MRSA is resistant to several antibiotics, so these antibiotics are less effective against it and fewer treatments work.

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20 · Part 1 Extended Drugs [5]

Mark scheme

(a) **[2]**

- (980 − 600) ÷ 980 × 100 or 380 ÷ 980 × 100 (1)
- 38.8 (%) (1)

ALLOW 38.78 (%) or 38.776 (%) for the second mark

REJECT 39 (%) or 38.7 (%) for the second mark

ECF correct percentage calculation from a wrong subtraction

(b) **[1]**

- as the number of prescriptions decreased, the percentage of infections caused by MRSA also decreased (1)

ALLOW positive correlation / both decreased

(c) **[1]**

- any one of: more hand washing by staff and visitors / cleaning surfaces and equipment more often / keeping patients with MRSA away from other patients (isolation) / testing patients for MRSA on arrival (1)

ALLOW any other sensible measure that reduces the spread of bacteria between patients, e.g. sterilising equipment, wearing gloves or aprons

(d) **[1]**

- antibiotics are still needed to treat serious / essential bacterial infections (1)

ALLOW some patients would die or become very ill without them

Worked solution

(a) The decrease in prescriptions is 980 − 600 = 380 per 1000 people. Percentage decrease = 380 ÷ 980 × 100 = 38.78%, which is 38.8% to three significant figures. Check: 38.8% of 980 is about 380. (b) As the number of prescriptions fell from 980 to 600, the percentage of infections caused by MRSA fell from 42% to 22%, so the two decreased together. (c) MRSA can also be reduced by stopping it spreading between patients, for example by hand washing, cleaning surfaces or isolating infected patients. (d) Antibiotics are still essential for treating serious bacterial infections; the aim is to use them only when they are needed, not to stop using them.

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